
Transfer Learning on the Oxford-IIIT Pet Dataset

Xianhan Li

KTH Royal Institute of Technology
xianhan@kth.se

Dehan Wang

KTH Royal Institute of Technology
dehanw@kth.se

Jinbin Liu

KTH Royal Institute of Technology
jinbin@kth.se

Yiqun Cui

KTH Royal Institute of Technology
yiqunc@kth.se

Abstract

This project studies transfer learning on the Oxford-IIIT Pet Dataset using an ImageNet-pretrained ResNet34 model. The E-level experiments include binary cat-versus-dog classification, 37-class breed classification, linear probing, fine-tuning strategies, and evaluations under limited and imbalanced data settings. Results show that pretrained features provide strong baseline performance. Further fine-tuning improves performance, while gradual unfreezing shows better robustness when labeled data is limited.

The B/A-level extension investigates semi-supervised pseudo-labeling. First, a teacher model generates high-confidence pseudo-labels for unlabeled training images. Then, a student model is trained using both labeled and pseudo-labeled data. Pseudo-labeling improves test accuracy from 89.07% to 89.59% in the 50% labeled setting, but shows almost no improvement with 10% and 1% labeled data.

1 Introduction

In the E-level basic project, pretrained ResNet models are applied to the Oxford-IIIT Pet dataset for both binary cat-versus-dog classification and 37-class breed classification. The study investigates the effectiveness of transfer learning through linear probing and different fine-tuning strategies, including simultaneous fine-tuning and gradual unfreezing. Additional experiments are conducted under limited labeled data and imbalanced class distributions to evaluate the effects of data augmentation, regularization, and imbalance-handling techniques. The results demonstrate that pretrained representations provide strong baseline performance, while fine-tuning further improves classification accuracy, especially under limited-data conditions.

In the B/A-level extension, pseudo-labeling is used to test whether unlabeled images can improve performance when labeled data is limited. First, a teacher model is trained on the labeled subset, and then it assigns pseudo-labels to the remaining unlabeled images. Only high-confidence predictions are retained, and a student model is trained using both labeled and pseudo-labeled data. We compare this method with supervised baselines under 100%, 50%, 10%, and 1% labeled-data settings. The results indicate that pseudo-labeling is only helpful when the teacher model is reliable and can generate a sufficient number of high-confidence pseudo-labels.

2 Related Work

Semi-supervised learning aims to improve model performance when labeled data is limited by using additional unlabeled samples. Pseudo-labeling is a simple and widely used approach in this context: a model trained on labeled data predicts labels for unlabeled samples, and high-confidence predictions

are then added back to the training set. Lee [2013] proposes pseudo-labeling as an efficient semi-supervised learning method for deep neural networks, which directly inspires our B/A-level extension. Our experiments use the Oxford-IIIT Pet Dataset introduced by Parkhi et al. [2012], a fine-grained benchmark containing 37 cat and dog breeds. We apply pseudo-labeling to this transfer learning task and analyze whether it can improve over supervised baselines under 50%, 10%, and 1% labeled-data settings.

3 Dataset and Experimental Setup

In the E-level basic project, experiments are conducted using the Oxford-IIIT Pet dataset, which contains 37 pet breeds, including 12 cat breeds and 25 dog breeds. The dataset is divided into 3131 training images, 549 validation images, and 3669 test images. To investigate transfer learning under different conditions, additional experiments are performed using limited training subsets (100%, 10%, and 1% of the training data) as well as an imbalanced dataset in which cat-breed samples are reduced to 20% of their original size.

The B/A-level extension uses the same Oxford-IIIT Pet dataset and the same training, validation, and test split as the E-level experiments. The main difference is that the training set is divided into labeled and unlabeled subsets for semi-supervised learning. We evaluate 100%, 50%, 10%, and 1% labeled-data settings. For each setting, the labeled subset is selected in a stratified manner, and the remaining training images are treated as unlabeled data for pseudo-labeling. The validation and test sets remain unchanged, with the test set used only for final evaluation.

4 E-Level Basic Transfer Learning Experiments

4.1 Methods and Experiments

4.1.1 Binary Classification

As an initial sanity check, the pretrained ResNet34 model is adapted for binary cat-versus-dog classification by replacing the final layer with a two-class classifier while keeping all convolutional layers frozen. The model achieves 99.10% test accuracy and a macro F1-score of 98.97%, indicating that pretrained ImageNet features are highly effective for simple animal classification tasks.

4.1.2 Linear Probing

To evaluate the quality of pretrained representations, linear probing is performed on the 37-class breed classification task by training only the final classification layer of a frozen ResNet34 model. The model achieves 90.08% test accuracy and a macro F1-score of 89.96%, demonstrating strong transferability of ImageNet-pretrained features to the Oxford-IIIT Pet dataset.

4.1.3 Fine-tuning Strategies

Two fine-tuning strategies are compared for the 37-class breed classification task. Strategy 1 simultaneously unfreezes the last (l) ResNet layer groups together with the classification layer, while Strategy 2 gradually unfreezes additional layers during training. In Strategy 1, the highest validation accuracy is obtained with ($l = 3$), reaching 93.62%, although the best test accuracy is achieved with ($l = 2$) at 92.15%. This indicates that increasing the number of trainable layers does not always improve generalization and may lead to overfitting or disruption of pretrained representations. Detailed results for different values of l are provided in the Appendix. Strategy 2 achieves comparable performance, with 93.08% validation accuracy and 91.11% test accuracy, but requires longer training time due to its multi-stage optimization process.

Table 1: Comparison of Fine-Tuning Strategies

Strategy	Validation Accuracy(%)	Test Accuracy(%)	Train time(s)
Strategy 1 ($l=3$)	93.62	90.98	45.50
Strategy 2	93.08	91.11	81.80

4.1.4 Limited Data Experiments

To investigate the effect of limited labeled data, stratified subsets containing 100%, 10%, and 1% of the training set are used. Experiments are performed using different combinations of data augmentation and L2 regularization. Model configurations are selected according to validation accuracy, while final performance is evaluated on the test set.

The results show that the classification performance decreases substantially as the amount of labeled data is reduced. Using the best validation-selected configuration for each data fraction, the model achieves 90.41% test accuracy with the full training set, 81.74% with 10% of the training data, and 23.88% with only 1% of the data. Data augmentation is most beneficial in the 10% setting, suggesting that augmentation becomes more useful when labeled data is limited. However, the 1% setting remains highly challenging even with augmentation and regularization. Detailed results are provided in the Appendix.

4.1.5 Strategy Comparison Under Limited Data

The two fine-tuning strategies are further compared using training fractions of 100%, 10%, and 1%. With the full training dataset, Strategy 1 achieves slightly better performance, reaching 91.82% test accuracy compared to 90.60% for Strategy 2. However, as the amounts of labeled data decrease, gradual unfreezing becomes increasingly advantageous. Using training fractions of 10% and 1%, Strategy 2 achieves higher validation accuracy than Strategy 1. With 10% of the training data, Strategy 2 achieves 82.01% test accuracy versus 80.43% for Strategy 1, while in the 1% setting the gap increases substantially to 41.78% versus 20.22%. These results suggest that gradual unfreezing preserves pretrained representations more effectively and reduces overfitting when labeled data is extremely limited. Detailed results are provided in the Appendix.

4.1.6 Imbalanced Data Experiments

To simulate class imbalance, the number of training samples for all cat breeds is reduced to 20% of their original size while all dog breed samples remain unchanged. Three approaches are evaluated: training without compensation, weighted cross-entropy loss, and oversampling of minority classes. Performance is measured using test accuracy together with macro and weighted F1 scores.

Results show that class imbalance mainly reduces minority-class performance. Training without compensation achieves the best overall performance, reaching 87.79% test accuracy, an 87.55 macro F1 score and 87.66 weighted F1 score. Oversampling produces similar results, while weighted cross-entropy results in slightly lower performance. These findings suggest that pretrained ImageNet representations already provide some robustness against moderate class imbalance, and that imbalance compensation strategies provide only limited benefits under the current experimental setup. Detailed results are provided in the Appendix.

4.2 Conclusion

In this section, transfer learning with pretrained ResNet34 models is investigated on the Oxford-IIIT Pet dataset. Experimental results show that pretrained ImageNet representations already provide strong performance through linear probing, while fine-tuning further improves classification accuracy. Additional experiments demonstrate that data augmentation becomes more beneficial as labeled data decreases and that gradual unfreezing provides better robustness in extremely low-data settings. Finally, pretrained representations remain relatively robust under moderate class imbalance, although minority-class performance is still reduced.

5 B/A-Level Extension: Semi-Supervised Pseudo-Labeling

5.1 Methods and Experiments

5.1.1 Pseudo-Labeling Setup

The B/A-level extension investigates whether unlabeled training images can improve performance when only a limited fraction of labels is available. We employ a pseudo-labeling pipeline inspired by Lee [Lee, 2013]. For each labeled-data fraction, a teacher model is first trained using only the labeled

subset. The remaining training images are treated as unlabeled data. The teacher model then predicts class probabilities for these images, and only predictions whose maximum softmax confidence is above a threshold τ are kept as pseudo-labels. Finally, a student model is trained on the union of the labeled subset and the selected pseudo-labeled samples.

We evaluate four labeled-data settings: 100%, 50%, 10%, and 1%. The 100% setting has no unlabeled subset and is used as the full supervised reference. For the 50%, 10%, and 1% settings, the labeled subset is selected in a stratified manner, while the remaining training samples are used for pseudo-label generation. The teacher and student use the same fine-tuning strategy within each fraction: Strategy 1 with $l = 3$ is used for 100% and 50%, while gradual unfreezing is used for 10% and 1%, following the limited-data results from the basic experiments.

5.1.2 Supervised Baselines under Different Label Fractions

Before applying pseudo-labeling, we first train supervised baselines under different labeled-data fractions. These baselines are used as the direct comparison targets for pseudo-labeling, because each pseudo-labeling model should be compared with a supervised model trained using the same amount of labeled data.

Table 2: Supervised baselines under different labeled-data fractions.

Fraction	Strategy	Labels	Test Acc.(%)	Macro F1(%)
100%	S1, $l = 3$	3131	90.32	90.16
50%	S1, $l = 3$	1549	89.07	88.67
10%	S2	296	81.14	80.76
1%	S2	37	40.75	37.52

The supervised results show a clear drop in performance as the labeled subset becomes smaller. The full-data model achieves 90.32% test accuracy and 90.16% macro F1, while the 1% setting drops to 40.75% test accuracy and 37.52% macro F1. This confirms that limited labeled data is a meaningful setting for evaluating whether unlabeled data can provide additional useful supervision.

5.1.3 Pseudo-Labeling Main Results

We then apply pseudo-labeling with a confidence threshold of $\tau = 0.9$. Table 3 compares the student model with the supervised baseline under the same labeled-data fraction.

Table 3: Pseudo-labeling results compared with supervised baselines.

Fraction	Supervised Acc.(%)	Pseudo Acc.(%)	Δ Acc.(%)	Pseudo Labels	Accept Rate
100%	90.32	90.32	+0.00	0	0.0%
50%	89.07	89.59	+0.52	559	35.3%
10%	81.14	81.11	-0.03	155	5.5%
1%	40.75	39.96	-0.79	1	0.0%

The results show that pseudo-labeling is only clearly beneficial in the 50% labeled setting. Here, the teacher accepts 559 pseudo-labeled samples with a 35.3% acceptance rate, and the student improves test accuracy from 89.07% to 89.59%. This suggests that the teacher is reliable enough to generate useful additional supervision.

By contrast, pseudo-labeling gives almost no gain in the 10% setting and slightly hurts performance in the 1% setting. The accepted pseudo-labeled samples drop to 155 at 10% and only 1 at 1%, showing that the teacher can no longer provide enough useful high-confidence samples. Overall, simple pseudo-labeling depends on both teacher quality and pseudo-label coverage: it can help in a moderately limited setting, but becomes ineffective when labeled data is severely limited.

5.1.4 Confidence Threshold Ablation

To analyze the effect of the confidence threshold, we conduct an ablation study at the 10% labeled-data setting using $\tau \in \{0.7, 0.8, 0.9, 0.95\}$. For this ablation, a separate 10% supervised teacher is trained and then fixed while varying the confidence threshold. Therefore, all Δ Acc. values in Table 4

are computed relative to this teacher baseline of 82.04%, rather than the 10% supervised baseline in the main comparison table.

Table 4: Confidence threshold ablation at 10% labeled data.

Setting	Pseudo Labels	Accept Rate	Test Acc.(%)	Δ Acc.(%)
Teacher baseline	–	–	82.04	0.00
$\tau = 0.70$	467	16.5%	81.68	-0.35
$\tau = 0.80$	172	6.1%	79.97	-2.07
$\tau = 0.90$	26	0.9%	81.63	-0.41
$\tau = 0.95$	1	0.0%	81.19	-0.84

The results show a clear trade-off between pseudo-label quantity and reliability. Lowering the threshold accepts more samples, but this does not necessarily improve the student model. For example, $\tau = 0.7$ accepts 467 pseudo-labeled samples with a 16.5% acceptance rate, but the test accuracy is still below the teacher baseline. This suggests that some accepted pseudo-labels may be noisy or not sufficiently useful for training.

Higher thresholds produce more reliable pseudo-labels but accept too few samples. With $\tau = 0.9$ and $\tau = 0.95$, only 26 and 1 samples are accepted, respectively, so the additional supervision is too small to improve performance. Since none of the thresholds outperforms the teacher baseline, the limitation is not only the threshold choice. Simple pseudo-labeling requires both reliable predictions and enough accepted samples, which are difficult to obtain in the 10% labeled setting.

5.1.5 Failure Analysis

We further analyze the 10% labeled setting to understand why pseudo-labeling yields only a marginal change in overall performance despite achieving high pseudo-label accuracy. As shown in Table 5, the overall test accuracy remains almost unchanged, from 81.14% for the supervised baseline to 81.11% for pseudo-labeling. However, the class-group behavior changes more noticeably. Cat-breed macro F1 increases from 71.77% to 74.92%, while dog-breed macro F1 decreases from 85.79% to 84.87%. This indicates that pseudo-labeling does not uniformly improve all classes, but changes the distribution of errors across different class groups.

Table 5: Failure analysis at the 10% labeled setting.

Metric	Supervised	Pseudo-labeling	Difference
Test accuracy	81.14	81.11	-0.03
Macro F1	80.76	80.94	+0.18
Cat-breed macro F1	71.77	74.92	+3.15
Dog-breed macro F1	85.79	84.87	-0.92
Accepted pseudo-labels	–	155	–
Pseudo-label accuracy	–	98.71	–

Figure 1 visualizes the group-level F1 scores. Although the overall accuracy is almost unchanged, the cat-breed macro F1 improves noticeably. This suggests that pseudo-labeling affects the class-wise decision boundaries even when the aggregate accuracy does not clearly improve.

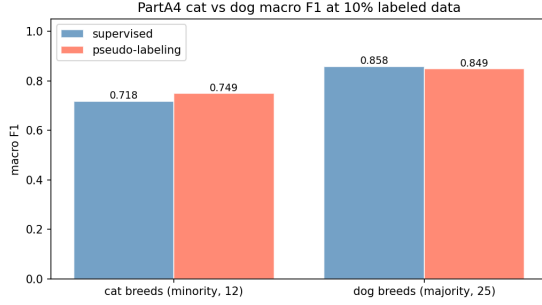


Figure 1: Cat-breed and dog-breed macro F1 comparison at the 10% labeled setting.

To further inspect the error patterns, we summarize the most frequent confused class pairs in Table 10. Most errors occur between visually similar breeds, such as Siamese and Birman, Abyssinian and Bengal, Russian Blue and British Shorthair, and American Pit Bull Terrier and Staffordshire Bull Terrier. The full confusion class pairs are provided in Table 10.

The top confused pairs show that pseudo-labeling does not remove the main fine-grained confusions. Some errors are reduced, but others become more frequent. For example, the confusion from Russian Blue to British Shorthair increases from 22 to 36 cases, and American Pit Bull Terrier to Staffordshire Bull Terrier increases from 23 to 33 cases. This suggests that the student model can inherit the teacher’s uncertainty between visually similar breeds. Therefore, pseudo-labeling is not limited only by pseudo-label accuracy. Even though the accepted pseudo-labels have high estimated accuracy, the accepted set is small and likely biased toward easier examples, while difficult class boundaries remain underrepresented.

5.2 Conclusion

The investigation into semi-supervised pseudo-labeling reveals that leveraging unlabeled data provides nuanced, context-dependent benefits rather than a universal performance boost. The efficacy of the pseudo-labeling pipeline is strictly bound by a delicate trade-off between teacher model reliability and pseudo-label coverage.

The investigation into semi-supervised pseudo-labeling reveals that leveraging unlabeled data provides nuanced, context-dependent benefits. In a moderately data-limited regime, pseudo-labeling yields a clear performance gain because the teacher model maintains a high enough baseline proficiency to confidently and accurately annotate a substantial portion of the unlabeled pool, injecting valuable additional supervision into the student model.

However, as labeled data becomes severely scarce, the limitations of simple threshold-based pseudo-labeling become starkly apparent. In high-scarcity regimes, the teacher model’s weak generalization renders the confidence threshold ineffective: lowering it introduces confirmation bias through noisy labels, while raising it yields a minuscule subset that fails to provide a meaningful training signal. Furthermore, pseudo-labeling alters decision boundaries unevenly across categories and fails to resolve fine-grained, visually similar confusions. This occurs because high-confidence selections are inherently skewed toward structurally easy examples, leaving ambiguous and critical class boundaries underrepresented and unrefined. Consequently, vanilla pseudo-labeling proves insufficient for low-resource settings, motivating the future exploration of more robust paradigms like consistency regularization or dynamic, class-adaptive thresholding.

6 Limitations and Future Work

The current extension has two main limitations. Firstly, each experiment was run only once, so small gains such as the 0.52 percentage-point improvement at 50% labeled data should be interpreted cautiously and verified with multiple random seeds. Secondly, the threshold ablation was only performed at 10% labeled data. Since pseudo-labeling works best at 50%, future work should also test threshold sensitivity in that setting.

References

- Dong-Hyun Lee. Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks. In *Workshop on Challenges in Representation Learning, ICML*, 2013.
- Omkar M. Parkhi, Andrea Vedaldi, Andrew Zisserman, and C. V. Jawahar. Cats and dogs. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2012.

A Additional Results

A.1 Results Table

Table 6: Strategy 1 Performance Under Different l

l	Validation Accuracy(%)	Test Accuracy(%)	Train time(s)
1	92.71	91.63	44.80
2	92.35	92.15	45.90
3	93.62	90.98	45.50
4	92.53	90.38	45.70
5	92.53	91.69	45.90

Table 7: Best validation-selected configuration for each labeled data fraction.

Fraction	Augment	Weight Decay	Validation Accuracy(%)	Test Accuracy(%)
100%	True	0	92.90	90.41
10%	True	0	80.87	81.74
1%	False	0	23.86	23.88

Table 8: Comparison between Strategy 1 and Strategy 2 under different labeled data fractions.

Fraction	Strategy	Validation Accuracy(%)	Test Accuracy(%)
100%	S1	92.71	91.82
100%	S2	92.53	90.60
10%	S1	76.68	80.43
10%	S2	80.69	82.01
1%	S1	20.95	20.22
1%	S2	40.80	41.78

Table 9: Performance under imbalanced training data using different compensation strategies.

Method	Test Accuracy(%)	Macro F1	Weighted F1
No compensation	87.79	87.55	87.66
Weighted CE	86.94	86.78	86.85
Oversampling	87.60	87.37	87.44

Table 10: Confused class pairs under 10% labeled setting

Method	True	Predicted	Count
Supervised	Siamese	Birman	35
	Abyssinian	Bengal	25
	Bengal	Egyptian Mau	23
	American Pit Bull Terrier	Staffordshire Bull Terrier	23
	Russian Blue	British Shorthair	22
	Maine Coon	Persian	21
	Russian Blue	Egyptian Mau	19
	Ragdoll	Birman	19
	Ragdoll	Persian	18
	English Cocker Spaniel	English Setter	18
Pseudo-labeling	Russian Blue	British Shorthair	36
	American Pit Bull Terrier	Staffordshire Bull Terrier	33
	Maine Coon	Bengal	22
	Abyssinian	Bengal	22
	Ragdoll	Birman	19
	Birman	Siamese	19
	English Setter	English Cocker Spaniel	17
	Miniature Pinscher	American Pit Bull Terrier	12
	Chihuahua	Sphynx	12
	Birman	Ragdoll	12

A.2 Code Availability

The source code for this project is publicly available on GitHub:

<https://github.com/12013001/DD2424-Project-Group25>